Sampling and Reconstruction

ECSE 313 Group 16 Lab 4 Report

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1 Sampling and Reconstruction Using Sample-and-Hold

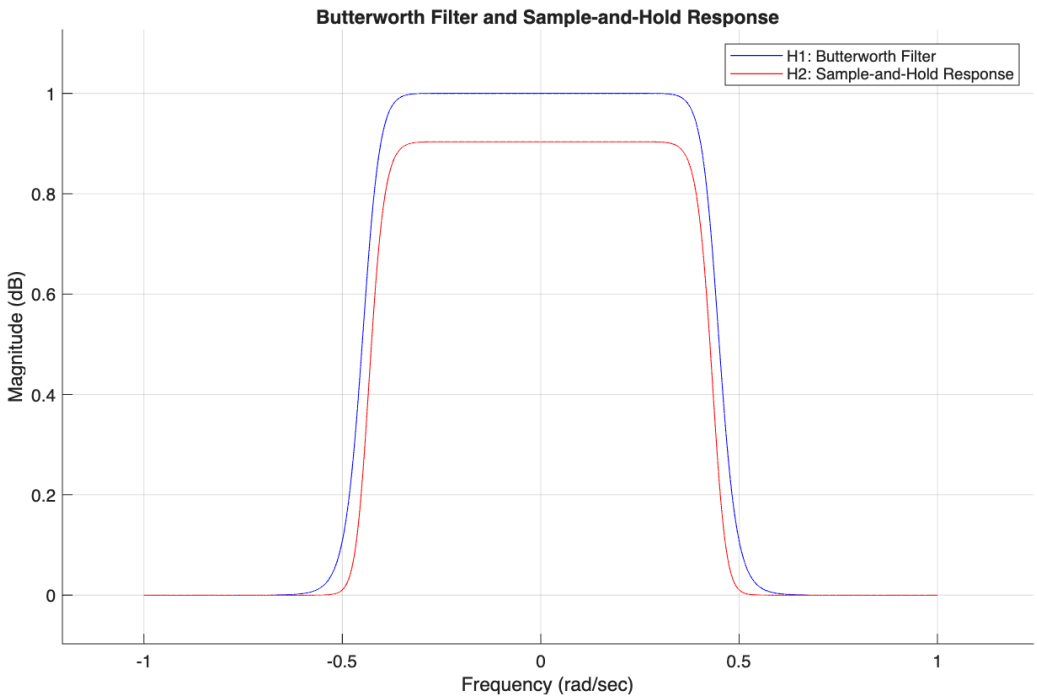

Figure 1: Magnitude Response of Butterworth Filter and Sample-and-Hold Response

The magnitude of the Butterforth filter is greater than that with the sample-and-hold device. The magnitude response of the sample-and-hold acts as a low pass filter. For high quality audio CD players, a low pass filter smooths the signal and improves audio clarity.

2 Sampling and Reconstruction Using An Impulse Generator

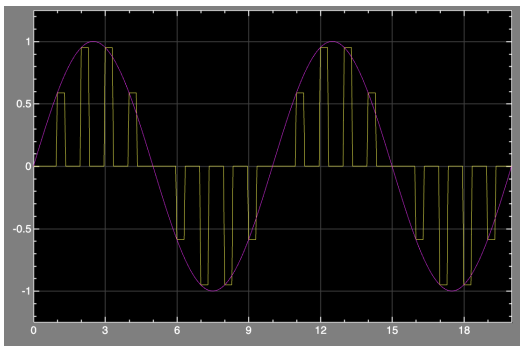

Figure 2: Input/Output Signals at 0.3 Sec Pulse

The aliasing in Figure 2 is represented by the yellow curve.

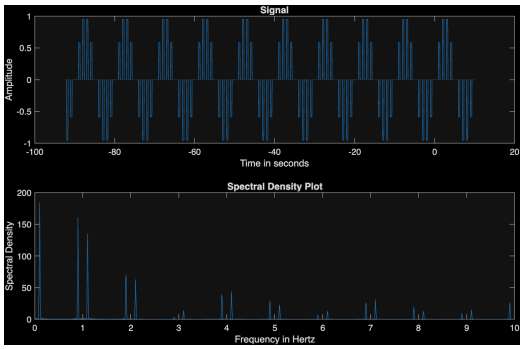

Figure 3: Output Signal and Frequency Spectrum at 0.3 Sec Pulse

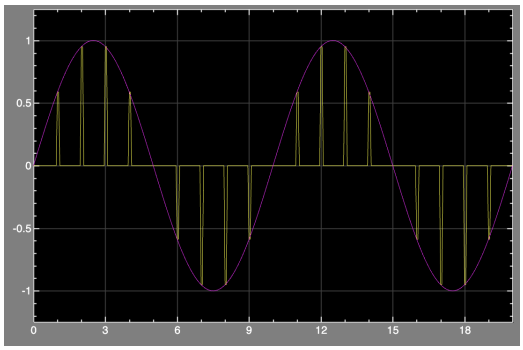

Figure 4: Input/Output Signals at 0.1 Sec Pulse

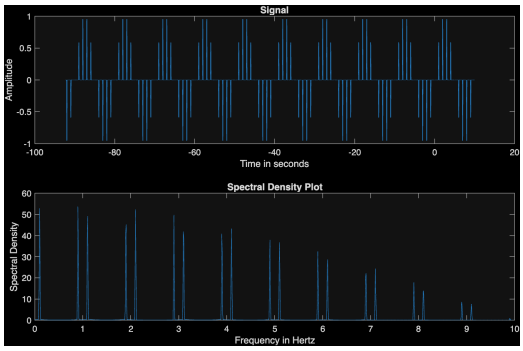

Figure 5: Output Signal and Frequency Spectrum at 0.1 Sec Pulse

The aliasing curves in Figure 4 have a smaller width than those in Figure 2. Additionally, the spectral density in Figure 5 has a lower magnitude intensity than that in Figure 3.

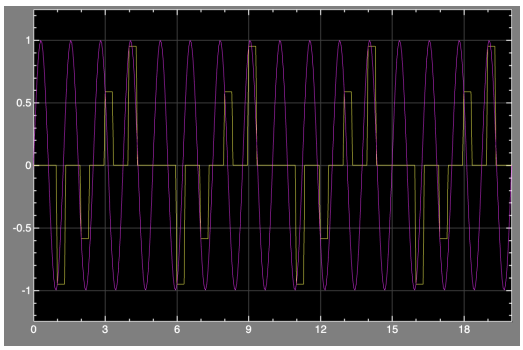

Figure 6: Input/Output Signals at 0.1 Sec Pulse

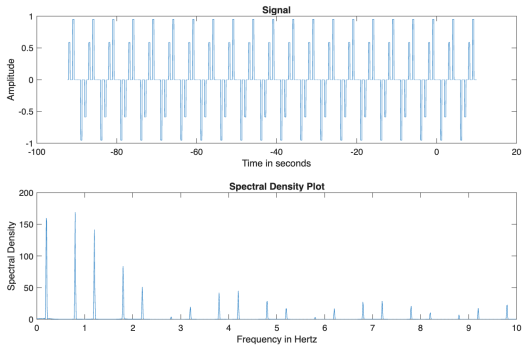

Figure 7: Output Signal and Frequency Spectrum at 0.3 Sec Pulse with 0.8 Hz

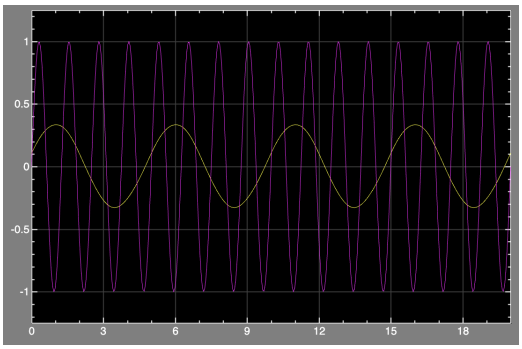

Figure 8: Input/Output Signals with Butterworth Filter

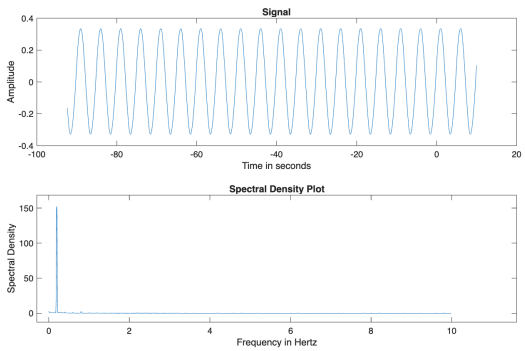

Figure 9: Output Signal and Frequency Spectrum with Butterworth Filter

3 Sampling and Reconstruction with Sample and Hold

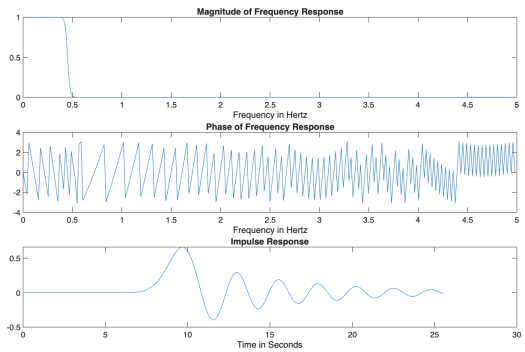

Figure 10: Sample-and-Hold With Two Scopes

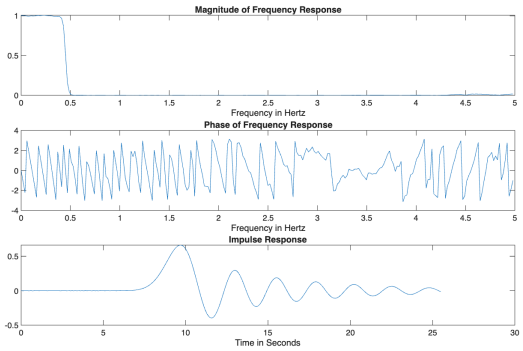

Figure 11: Sample-and-Hold With Two Scopes

4 Discrete-Time Interpolation

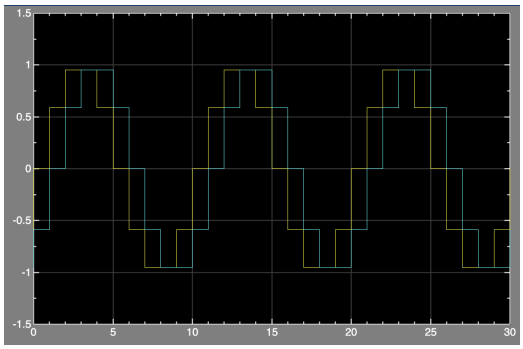

Figure 12: Discrete Time Interpolator Scope

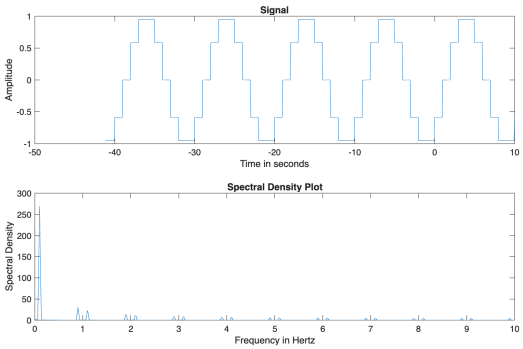

Figure 13: Signal.c and Frequency Spectrum

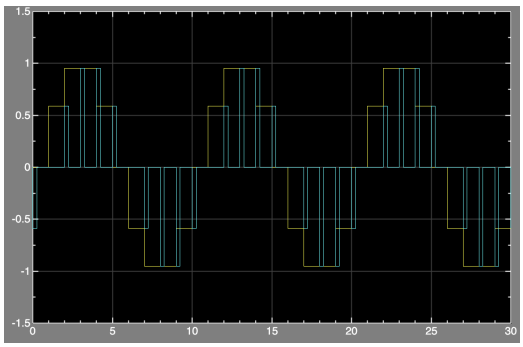

Figure 14: Discrete Time Interpolator Scope With 4 Hz

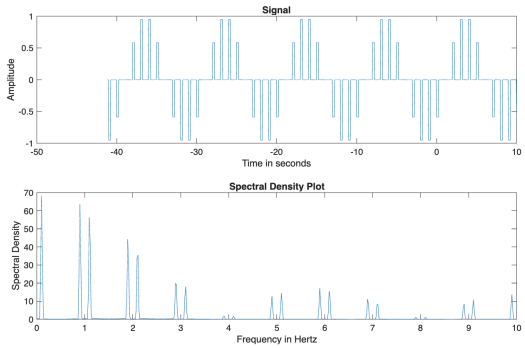

Figure 15: Signal.c and Frequency Spectrum With 4 Hz

5 Discrete-Time Decimation